Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2025

Course: B.Tech
Code: 103402

Subject: Electrical Machines – II

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

✓***Q.1*** ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) For a 3-phase uniformly distributed and narrow-spread winding, the distribution factor is
- (i) 0.9
- (ii) 0.827
- (iii) 0.9555
- (iv) 0.97

(b) Compared to concentrated winding, a distributed winding mainly helps in
- (i) increasing copper loss
- (ii) increasing leakage flux
- (iii) reducing harmonics
- (iv) reducing efficiency

(c) A pulsating magnetic field can be resolved into
- (i) two unequal rotating fields
- (ii) two equal and opposite rotating fields
- (iii) a single rotating field
- (iv) a stationary field only

(d) Increasing rotor resistance of a slip-ring induction motor
- (i) increases starting torque
- (ii) increases efficiency
- (iii) reduces starting torque
- (iv) reduces slip at maximum torque

(e) In case of 3-phase IM, shaft power is 2700 W and mechanical losses are 180 W. At a slip of 4%, the rotor ohmic losses are
- (i) 115.2 W
- (ii) 108 W
- (iii) 120 W
- (iv) 105 W

(f) Most common application of 3-phase induction generator can be in
- (i) steam-power station
- (ii) wind-power station
- (iii) hydro-power station
- (iv) nuclear-power station

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(g) In a 4-pole, 3-phase, 50 Hz induction machine, the slip-rings of the machine are open-circuited. The frequency of the voltage across slip rings is 75 Hz. The rotor is driven at speed of
- (i) 750 rpm
- (ii) 2250 rpm
- (iii) 1500 rpm
- (iv) 1000 rpm

(h) Single-phase induction motors are not self-starting because
- (i) rotor is squirrel cage type
- (ii) supply frequency is low
- (iii) starting torque is zero
- (iv) stator has distributed winding

(i) A under excited synchronous motor will operate at ________ power factor.
- (i) unity
- (ii) leading
- (iii) lagging
- (iv) leading or lagging

(j) Which of the following condition is true for a salient pole synchronous machine?
- (i) $X_d = X_q$
- (ii) $X_d < X_q$
- (iii) $X_d > X_q$
- (iv) None of these

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Q.2 (a) Derive the expression for distribution factor. What are the advantages of short-pitch winding? [7]

(b) A 3-phase, 50 Hz, 2-pole, star-connected turbo alternator has 54 slots with 4 conductors per slot. The pitch of the coil is 2 slots less than the pole pitch. If the machine gives 3300 V between lines on open circuit with sinusoidal flux distribution, determine the useful flux per pole. [7]

Q.3 (a) Using mathematical analysis, show how a pulsating magnetic field can be resolved into two rotating magnetic fields. [7]

(b) Describe the effect of spatial displacement of windings on magnetic field production. [7]

Q.4 (a) Derive the torque–slip characteristic of an induction motor and explain the condition for maximum torque. [7]

(b) A 4-pole, 50 Hz, 3-phase induction motor has rotor resistance and standstill rotor reactance of 0.04 Ω and 0.16 Ω per phase respectively. Calculate the value of the external rotor resistance per phase to be inserted to obtain 70% of maximum torque at starting. [7]

Q.5 (a) Explain the no-load test and blocked rotor test of an induction motor. How are the parameters determined from these tests? [7]

(b) A 4-pole, 3300 V, 50 Hz induction motor runs at rated frequency and voltage. The frequency of the rotor currents 2.5 Hz. Find the per unit slip and the running speed. [7]

Q.6 (a) Describe the construction and working of split-phase induction motors. [7]

(b) Explain the equivalent circuit of a single-phase induction motor and determination of its parameters. [7]

Q.7 (a) Explain the effect of varying excitation on armature current and power factor in a synchronous motor with appropriate phasor diagram. Draw V-curves and state their significance. [7]

(b) Two exactly similar 3000 kVA synchronous generator operate in parallel. The governor of the first machine is such that the frequency drops uniformly from 50 Hz on load to 48 Hz on full load. The corresponding uniform speed drop of second machine is from 50 Hz to 47.5 Hz. How will the two machines share a load of 5000 kW? [7]

Q.8 (a) Explain the two reaction theory applicable to salient pole synchronous machine. [7]

(b) A 15 kW, 3-phase, star connected synchronous motor operating on full load from infinite busbars, has its excitation so adjusted that the power factor is 0.8 lagging. Load being constant, excitation is now increased by 25%. Synchronous reactance is 1.0 per unit. Find the new power factor. [7]

Q.9 Write short note on ***any two*** of the following: **[7 x 2 = 14]**

(a) Stepper motor

(b) Brushless DC motor

(c) Hysteresis motor

(d) Parallel operation of 3-phase alternator

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